For questions 1-5, consider a particle in a one-dimensional infinite square well with width $a$, centered at $a/2$. The normalized energy eigenstate wave functions are $\psi_n(x)$ with energies $E_n (n = 1$ is the ground state). The particle starts in a state given by

$$\psi(x, t = 0) = \sqrt{\frac{4}{5}}\,\psi_1(x) + \sqrt{\frac{1}{5}}\,\psi_2(x).$$

1. What is the maximum value that could be measured for the energy of this system?

A. $\frac{4}{5}$
B. $E_1$
C. $E_2$
D. $\frac{4}{5}E_1 + \frac{1}{5}E_2$
E. Any $E_n$ could be measured.

2. Suppose instead you measure energy of the system above, $\left(\sqrt{\frac{4}{5}}\,\psi_1(x) + \sqrt{\frac{1}{5}}\,\psi_2(x)\right)$, and this time you find the most probable value for the energy. What is the normalized state of the system after this measurement?

A. $\sqrt{\frac{4}{5}}\,\psi_1(x)$
B. $\sqrt{\frac{1}{5}}\,\psi_2(x)$
C. $\psi_1(x)$
D. $\psi_2(x)$
E. $\sqrt{\frac{4}{5}}\,\psi_1(x) + \sqrt{\frac{1}{5}}\,\psi_2(x)$

Once again, we have a particle in a one-dimensional infinite square well starting in the state

$$\psi(x, t = 0) = \sqrt{\frac{4}{5}}\,\psi_1(x) + \sqrt{\frac{1}{5}}\,\psi_2(x).$$

3. You make a position measurement on the particle described above with high accuracy. After this position measurement, you immediately measure the energy. At this point, what value(s) could you get for the energy of the particle?

A. $\frac{4}{5}E_1 + \frac{1}{5}E_2$
B. $E_1$ only
C. Just $E_1$ or $E_2$
D. Any $E_n$
E. Any continuous energy $E$

4. Instead of an immediate energy measurement, consider waiting between the position and energy measurements. Would the probability of measuring $E_2$ depend on how long you wait between the position and energy measurements?

A. Yes
B. No

5. Choose the best explanation for your answer to the previous question from the choices below.

A. Over time the wave function of the particle will relax to the initial state.
B. The probability of measuring a particular energy is time-dependent.
C. The probability of measuring a particular energy is time-independent.
D. Although the probability of measuring a particular energy will vary in time, the expectation value of energy is time-independent.
E. The position measurement would not alter the energy of the system.

For questions 6-9, the normalized wave function for a particle in a one-dimensional infinite square well is given at time $t = 0$ by

$$\psi(x) = C_1\psi_1(x) + C_2\psi_2(x),$$

where $\psi_1(x)$ and $\psi_2(x)$ are distinct energy eigenfunctions, and the Hamiltonian is $\widehat{H}$.

6. Immediately after an energy measurement, what is the state of the system?

A. $C_1E_1\psi_1(x) + C_2E_2\psi_2(x)$
B. $E_1\psi_1(x) + E_2\psi_2(x)$
C. Undetermined, but it is either $E_1\psi_1(x)$ or $E_2\psi_2(x)$
D. Undetermined, but it is either $\psi_1(x)$ or $\psi_2(x)$

7. Immediately after an energy measurement on the state $\psi(x) = C_1\psi_1(x) + C_2\psi_2(x)$, the system is in the state $\widehat{H}\psi(x)$.

A. True
B. False

8. There is one definite energy $E$ associated with the state $\psi(x) = C_1\psi_1(x) + C_2\psi_2(x)$.

A. True
B. False

9. For this state, $\psi(x) = C_1\psi_1(x) + C_2\psi_2(x)$, there is a certain probability that an energy measurement will yield energy $E_2$.
If the constant $C_2$ is replaced with $C_2 e^{i\theta_2}$ (with $\theta_2$ a real number, $0 < \theta_2 < 2\pi$), what happens to the probability that an energy measurement will result in $E_2$?

A. The probability will not change
B. The probability will change
C. The probability depends on the particular value of $\theta_2$

10. A particle is in a one-dimensional infinite square well initially in the ground state. As time passes, which of the following statements describes the behavior of the wave function of the particle?

A. The real part will remain constant; the imaginary part is always zero.
B. The real part will oscillate with time; the imaginary part is always zero.
C. The real and imaginary parts will both oscillate with time
D. The real part will remain constant; the imaginary part will oscillate with time.
E. The real and imaginary parts will both remain constant.

11. What part(s) of the time dependent (in general complex) wave function $\psi(x,t)$ do you need in order to predict the probability of a particular outcome of an arbitrary position measurement?

A. Only the real part.
B. Only the imaginary part.
C. Both the real and imaginary parts.
D. Either the real or imaginary part.
E. The wave function does not contain information about the measurement outcome.

12. Choose the best explanation for your answer to the previous question from the choices below.

A. Physical observables like position are real.
B. Only the real part has position dependence and $\langle x \rangle$ has no imaginary part as it cancels out in $\psi^*(x,t)\psi(x,t)$.
C. If $\psi(x,t)$ were purely real there would be no time dependence in $|\ \psi(x,t)\ |^2$.

D. $|\psi(x,t)|^2$ can only be determined with both, because $\psi(x,t)$ might not be purely real.
E. The wave function for this system does not have an imaginary part.

13. Which of the following statements are always true for a particle in a one-dimensional infinite square well?

A. The energy expectation value is time-dependent only for energy eigenstates.
B. The energy expectation value is time-independent only for energy eigenstates.
C. The energy expectation value is always time-independent.
D. The energy expectation value is always time-dependent.
E. None of the above.

14. Which of the following statements are always true for a particle in a one-dimensional infinite square well?

A. The position expectation value is time-dependent for energy eigenstates.
B. The position expectation value is time-independent for energy eigenstates.
C. The position expectation value is always time-independent.
D. The position expectation value is always time-dependent.
E. None of the above.

For questions 15-17, consider a particle in a one-dimensional infinite square well, where the energy eigenstate $|\psi_n\rangle$corresponds to an energy of $E_n$.

15. If you start at time $t = 0$ with wave function $|\psi_1\rangle$, does the position probability density vary with time?

A. Yes
B. No
C. More information is needed to decide

16. If you start at time $t = 0$ with wave function $\frac{1}{\sqrt{2}}(|\psi_1\rangle+|\psi_2\rangle)$, does the position probability density vary with time?

A. Yes
B. No
C. More information is needed to decide

17. Of the choices given below, which state will have the fastest variation in time for the position probability density?

A. $|\psi_5\rangle$
B. $\frac{1}{\sqrt{2}}(|\psi_2\rangle+|\psi_3\rangle)$

C. $\frac{1}{\sqrt{2}}(|\psi_1\rangle - |\psi_4\rangle)$

D. $\sqrt{\frac{1}{5}}\,|\psi_1\rangle + \sqrt{\frac{4}{5}}\,|\psi_3\rangle$

E. All of these states have time-independent position probability densities.

18. Is the following statement true or false for all operators $\hat{Q}$? A system which is in an eigenstate of $\hat{Q}$ will stay in that state until disturbed by measurement.

A. True

B. False

19. Choose the best explanation for your answer to the previous question from the choices below.

A. The system remains in an eigenstate of operator $\hat{Q}$ until measurement of a non-commuting operator.

B. The expectation value of $\hat{Q}$ is time-independent.

C. The system will evolve and the state cannot be known.

D. The operator $\hat{Q}$ may not commute with the Hamiltonian or $\hat{Q}$ may have an explicit time dependence.

E. The operator $\hat{Q}$ may not commute with the position operator.

For questions 20-21, consider a spin-1/2 particle. The normalized eigenstates of $S_z$(the z-component of spin) are labeled as $|+\rangle$, and $|-\rangle$, with corresponding eigenvalues $\pm\hbar/2$. Consider a particle which starts in a quantum state given by

$$|\psi\rangle = \sqrt{\frac{1}{5}}\;|+\rangle + \sqrt{\frac{4}{5}}\;|-\rangle.$$

20. What is the maximum value that could be measured for $S_z$(the z-component of spin) in this system?

A. $\frac{4}{5}$

B. $-\hbar/2$

C. $+\hbar/2$

D. $\frac{1}{5}\,\hbar/2 + \frac{4}{5}\,(-\hbar/2)$

E. $\sqrt{\frac{1}{5}}\,\hbar/2 + \sqrt{\frac{4}{5}}\,(-\hbar/2)$

21. Suppose instead you measure $S_z$ of the system above, $\left(\sqrt{\frac{1}{5}}\ |+\rangle + \sqrt{\frac{4}{5}}\ |-\rangle\right)$ and this time you find the most probable value for the z-component of spin. What is then the normalized state of the system after this measurement?

A. $\sqrt{\frac{4}{5}}\ |-\rangle$

B. $\sqrt{\frac{1}{5}}\ |+\rangle$

C. $|-\rangle$

D. $|+\rangle$

E. $\sqrt{\frac{1}{5}}\ |+\rangle + \sqrt{\frac{4}{5}}\ |-\rangle$

For questions 22-24, consider a spin-1 particle sitting in a uniform magnetic field oriented in the $+z$direction. The Hamiltonian for this particle is proportional to $S_z$ (the z-component of spin). The normalized eigenstates of $S_z$ for this particle are labeled as $|-1\rangle$, $|0\rangle$ and $|1\rangle$, with corresponding eigenvalues $-\hbar$, $0$, and $+\hbar$. A particle starts in a quantum state given by

$$|\psi\rangle = \sqrt{\frac{4}{5}}\ |-1\rangle + \sqrt{\frac{1}{5}}\ |1\rangle.$$

22. Suppose you make a measurement of $S_x$ (the x-component of spin). Immediately after this measurement, you then measure $S_z$ (the z-component of spin). At this point, what value(s) could you get for the z-component of spin of the particle?

A. $\frac{4}{5}(-\hbar) + \frac{1}{5}(+\hbar)$

B. $-\hbar$ only

C. Just $-\hbar$ or $+\hbar$

D. $+\hbar$, $0$, or $-\hbar$

E. Any continuous angular momentum value between $-\hbar$ and $+\hbar$.

23. Instead of an immediate re-measurement of $S_z$, consider waiting between the $S_x$ and $S_z$ measurements. Would the probability of measuring $S_z$ to be $+\hbar$ depend on how long you wait between the measurements?

A. Yes

B. No

24. Choose the best explanation for your answer to the previous question from the choices below.

A. Over time the particle will relax to its initial state.
B. Because it is precessing in a magnetic field, the probabilities of $S_z$ change with time.
C. $S_z$ commutes with the Hamiltonian, so its probabilities are time-independent.
D. All possible measurement probabilities in quantum mechanics are time independent because you take the absolute square of the coefficient.
E. Once you measure $S_x$ the state is no longer in an eigenstate of the Hamiltonian.

For questions 25-28, the normalized wave function for a spin-1/2 particle at time $t = 0$ is

$$|\psi\rangle = C_1 |+\rangle + C_2 |-\rangle.$$

where $|+\rangle$ and $|-\rangle$ are eigenvectors of the $\hat{S}_z$ operator with eigenvalues $\pm\hbar/2$ respectively.

25. Immediately after a measurement of $S_z$, what is the state of the system?

A. $C_1 \frac{\hbar}{2} |+\rangle - C_2 \frac{\hbar}{2} |-\rangle$
B. $\frac{\hbar}{2} |+\rangle - \frac{\hbar}{2} |-\rangle$
C. Undetermined, but it is either $\frac{\hbar}{2} |+\rangle$or $-\frac{\hbar}{2} |-\rangle$
D. Undetermined, but it is either $|+\rangle$or $|-\rangle$

26. Immediately after a measurement of $S_z$ on the state $|\psi\rangle = C_1 |+\rangle + C_2 |-\rangle$, the system is in the state $\hat{S}_z |\psi\rangle$.

A. True
B. False

27. There exists a single definite value for the z-component of spin associated with the state $|\psi\rangle = C_1 |+\rangle + C_2 |-\rangle$.

A. True
B. False

28. For this state, $|\psi\rangle = C_1 |+\rangle + C_2 |-\rangle$, there is a certain probability that an $S_z$ measurement on this state $|\psi\rangle$ will yield a value $-\hbar/2$. If the constant $C_2$ is replaced with $C_2 e^{i\theta_2}$(with $\theta_2$a real number, $0 < \theta_2 < 2\pi$), what happens to the probability that a measurement of $S_z$ will result in $-\hbar/2$?

A. The probability will not change
B. The probability will change
C. The probability depends on the particular value of $\theta_2$

For questions 29-31, consider a system where we can measure two quantum mechanical observables, "color" (represented by the $\hat{C}$ operator) and "size" (represented by the $\hat{S}$ operator). The "color" operator has three eigenvalues (red, green, and blue) and three corresponding eigenstates ($|\ r\rangle$, $|\ g\rangle$, and $|\ b\rangle$). The "size" operator also has three eigenvalues (small, medium, and large) with three corresponding eigenstates ($|\ s\rangle$, $|\ m\rangle$, and $|\ l\rangle$). We wish to understand how color and size interact. Since we cannot see the "color" or "size" of our particles with our eyes, we build a lab with a ColorTron™ device which measures the "color" eigenvalue of a particle and a SizeUp™ device which measures the particle's "size" eigenvalue.

29. In the setup just described, an experimenter measures the color of particles and then immediately runs all particles which measured red through the ColorTron. What are the possible results of this second measurement?

A. The only possible measurement is red with a 100% probability of measurement.
B. Red, green, and blue can be measured with equal probabilities.
C. Red, green, and blue can be measured, but their associated probabilities of measurement cannot be determined from the information given.
D. There is not enough information to answer this question.

30. 1000 red particles are immediately run through the SizeUp which measures the small eigenvalue 200 times, the medium eigenvalue 300 times, and the large eigenvalue 500 times. Which one of the following could be a valid representation of the $|\ r\rangle$ state in the "size" basis?

A. $200\ |\ s\rangle + 300\ |\ m\rangle + 500\ |\ l\rangle$
B. $\frac{1}{10}(2\ |\ s\rangle + 3\ |\ m\rangle + 5\ |\ l\rangle)$
C. $\frac{1}{\sqrt{10}}\left(e^{i\alpha_1}\sqrt{2}\ |\ s\rangle + e^{i\alpha_2}\sqrt{3}\ |\ m\rangle + e^{i\alpha_3}\sqrt{5}\ |\ l\rangle\right)$
D. $\frac{1}{10}\left(e^{i\alpha}\left(\sqrt{2}\ |\ s\rangle + \sqrt{3}\ |\ m\rangle + \sqrt{5}\ |\ l\rangle\right)\right)$
E. None of the above.

31. 1000 medium particles are immediately run through the ColorTron and then immediately run through the SizeUp without looking at which colors were produced. If "color" and "size" commute, what can you say about the results from the SizeUp measurements?

A. All three “size” eigenvalues could be measured.
B. Most but not all of the particles will be measured as medium.
C. All particles will be measured as medium.
D. The results are probabilistic and cannot be predicted.
E. The “color” measurement will affect the “size” measurement.

32. True or False: Applying the Hamiltonian to any quantum state gives you information about how that state will evolve in time.

A. Always true.
B. True only in special cases.
C. Always false.

33. Choose the best explanation for your answer to the previous question from the choices below.

A. The Hamiltonian describes the time evolution of the system by the Time-Dependent Schrödinger’s Equation.
B. The Hamiltonian provides an oscillatory time dependence to $\psi(x)$.
C. The Hamiltonian only measures the energy at $t = 0$.
D. The Hamiltonian has no time dependence and only gives the energy eigenvalues at time $t > 0$.
E. The Hamiltonian collapses the system, making it impossible to determine the energies later.